DEPARTMENT OF MATHEMATICS
ALAGAPPA UNIVERSITY
(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the Third Cycle and Graded as Category-I University by MHRD-UGC)
KARAIKUDI - 630 003, TAMIL NADU, INDIA
www.alagappauniversity.ac.in

2017	2018	2019	2018	2019
A+ Grade by NAAC (CGPA: 3.64)	Graded as Category-1 & Granted Autonomy	NIRF Rank: 28	Swachh Campus Rank : 4	India Rank : 20 BRICS Rank : 104 Asia Rank : 216

M. Phil., MATHEMATICS
[Choice Based Credit System (CBCS)]
[For the candidates admitted from the academic year 2022-2023 onwards]

Panel of Members-Broad Based Board of Studies

Chairperson

Dr. N. Anbazhagan, Professor & Head, Department of Mathematics, Alagappa University, Karaikudi. Teaching Experience: 20 years, Research experience: 20 years, Area of Research: Stochastic Modeling, Data mining.

Foreign Experts

Dr. Rozaini Roslan, Professor, Department of Mathematics & Statistics,Faculty of Applied Sciences and Technology,Universiti Tun Hussein Onn Malaysia,Pagoh, Muar 84600, Malaysia., rozaini@uthm.edu.my . Working Experience: 20 Years, Research Experience: 20 Years, Area of Research: Fluid Mechanics, Heat and Mass Transfer, Nanofluids

Indian Experts

Dr. R. Uthayakumar, Professor and Head, Department of Mathematics, Gandhigram Rural Institute,Dindugal Teaching Experience: 24 years, Research experience: 24 Years, Area of Research: Fractal Theory, Operations Research, Inventory Management and Control

Dr. S. Muralisankar, Professor, Department of Mathematics, Madurai Kamaraj University, Madurai. Teaching Experience: 18 Years, Research experience: 18 Years, Area of Research: Fixed Point Theory,Fuzzy Functional Differential Equations, Stability analysis of Dynamical Systems

Members

Dr. J. Vimala, Assistant Professor, Department of Mathematics, Alagappa University, Karaikudi. Teaching Experience: 18 years, Research Experience: 15 years, Area of Research: Algebra –Lattice Theory, Fuzzy Algebra, Decision Theory and Soft computing.

Dr. R. Raja, Assistant Professor, Ramanujan Centre for Higher Mathematics, Alagappa University, Karaikudi. Teaching Experience: 11 Years, Research Experience: 10 years, Area of Research: Abstract & Fractional Differential Equations, Stability Analysis of Dynamical Systems, Neural Networks, Synchronization Theory, Mathematical Modeling and Population Systems, Genetic Regulatory Networks, Complex Dynamical Networks and Multi-Agent Systems.

Dr. B. Sundaravadivoo, Assistant Professor, Department of Mathematics, Alagappa University, Karaikudi. Teaching Experience: 19 years, Research Experience: 4 year, Area of Research: Abstract & Fractional Differential Equations, Control Theory, Mathematical Modelling and Perturbation Theory, Optimal Control.

Dr. S. Amutha, Assistant Professor, Ramanujan Centre for Higher Mathematics, Alagappa University, Karaikudi. Teaching Experience: 13 years, Research Experience: 13 years, Area of Research: Graph Theory, Domination Theory, Algorithmic Graph theory, Discrete Mathematics, Cryptography.

Dr. R. Jeyabalan, Assistant Professor, Department of Mathematics, Alagappa University, Karaikudi. Teaching Experience: 7 years, Research Experience: 7 years, Area of Research: Magic Labeling Graph Theory, Fuzzy Topology and Fuzzy Magic Labeling Graph Theory.

Dr. M. Mullai, Assistant Professor, Directorate of Distance Education, Alagappa University, Karaikudi. Teaching Experience: 18 years, Research Experience: 15 years, Area of Research: Algebra & Fuzzy Algebra, Operations Research, Mathematical Modelling, Neutrosophic sets (Neutrosophic Inventory, Neutrosophic Graph theory, Neutrosophic Optimization, Neutrosophic Adhoc networks)

Co opted Member from the Industry:

Mr. S. Gnanapandithan, Senior Manager ,Cognizant Technology Solutions, Coimbature, Robotic Process Automation Architect.

Alumni

Dr. A. Tamilselvan, Professor & Head, Department of Mathematics, Bharathidasan University, Tirchirapalli. Teaching Experience: 21 years, Research Experience: 21 years, Area of Research: Differential Equations, Numerical Analysis, Fractional Differential Equations, Finite Difference Methods, Finite Volume Methods.

Ex-officio Member

Dr. V.Sivakumar, Professor and Co-ordinator, Curriculum Development Cell, Distance Education, Alagappa University, Karaikudi Teaching Experience: 24 years, Research Experience: 17 years, Area of Research: Marketing Management , Agricultural Marketing , International Logistics , Agricultural Logistics and SCM , Consumer Research.

ALAGAPPA UNIVERSITY

DEPARTMENT OF MATHEMATICS

Karaikudi -630003, Tamil Nadu.

REGULATIONS AND SYLLABUS-(CBCS-University Department)

[For the candidates admitted from the Academic Year 2022 – 2023 onwards]

Name of the Department: **Department of Mathematics**

Name of the Subject Discipline: **Mathematics**

Programme of Level: **M.Phil.**

Duration for the Course: **Full Time** (One Year)

1. Choice-Based Credit System

A choice-Based Credit System is a flexible system of learning. This system allows students to gain knowledge at their own tempo. Students shall decide on electives from a wide range of elective courses offered by the University Departments in consultation with the Department committee. Students undergo additional courses and acquire more than the required number of credits. They can also adopt an inter-disciplinary and intra-disciplinary approach to learning, and make the best use of the expertise of available faculty.

2. Programme

"Programme" means a course of study leading to the award of a degree in a discipline.

3. Courses

'Course' is a component (a paper) of a programme. Each course offered by the Department is identified by a unique course code. A course contains lectures/ tutorials/laboratory /seminar /project /practical training/report writing /Viva-voce, etc or a combination of these, to meet effectively the teaching and learning needs.

4. Credits

The term "Credit" refers to the weightage given to a course, usually in relation to the instructional hours assigned to it. Normally in each of the courses credits will be assigned on the basis of the number of lectures/tutorial/laboratory and other forms of learning required to complete the course contents in a 15-week schedule. One credit is equal to one hour of lecture per week. For laboratory/field work one credit is equal to two hours.

5. Semesters

An Academic year is divided into two **Semesters.** In each semester, courses are offered in 15 teaching weeks and the remaining 5 weeks are to be utilized for conduct of examination and evaluation purposes. Each week has 30 working hours spread over 5 days a week.

6. Departmental committee

The Departmental Committee consists of the faculty of the Department. The Departmental Committee shall be responsible for admission to all the programmes offered by the Department including the conduct of entrance tests, verification of records, admission, and evaluation. The Departmental Committee determines the deliberation of courses and specifies the allocation of credits semester-wise and course-wise. For each course, it will also identify the number of credits for lectures, tutorials, practicals, seminars etc. The courses (Core/Discipline Specific Elective/Non-Major Elective) are designed by teachers and approved by the Departmental Committees. Courses approved by the Departmental Committees shall be approved by the Board of Studies/Broad Based Board of Studies. A teacher offering a course will also be responsible for maintaining attendance and performance sheets (CIA -I, CIA-II, assignments and seminar) of all the students registered for the course. The Non-major elective programme, MOOCs coordinator and Internship Mentor are responsible for submitting the performance sheet to the Head of the department. The Head of the Department consolidates all such performance sheets of courses pertaining to the programmes offered by the department. Then forward the same to be Controller of Examinations.

7. Programme General Objectives- (PGO) Minimum 6 objectives are required

PGO-1	To apply precise, logical reasoning to problem solving.
PGO-2	To provide comprehensive curriculum to groom the students.
PGO-3	To inculcate innovative skills, team work, ethical practices to face the society.
PGO-4	To stimulate the students for future research.
PGO-5	To identify the challenging problems and find solutions.
PGO-6	To develop a multi-disciplinary approach for solving problems through core courses.

8. Programme Specific Objectives-(PSO)- Minimum 6 objectives are required

PSO-1	To provide the student with pertinent information in the field of Mathematics.
PSO-2	To teach the student with a broad understanding of Mathematical and their interactions with the Equations.
PSO-3	To include methods of facilitating learning such as projects, group work and participative learning
PSO-4	To establish inter-disciplinarily between Mathematics and other subjects from Humanities and the Social Sciences.
PSO-5	To learn to apply Mathematics to real life situations and help in problem solving.
PSO-6	To qualify national level competitive exams like CSIR-NET/GATE etc.

9. Programme Outcome-(PO) - Minimum 6 objectives are required

PO-1	The students will learn to solve advanced mathematical equations theoretically/MATLAB.
PO-2	The students will know the concept of topological vector space and separation properties.
PO-3	The students will learn the fundamentals of carrying out a research as well as how to plan lessons, carry them out, and analyze the findings.
PO-4	The students will be able to acquire knowledge of product measures, Convolution and Distribution functions.
PO-5	The students will be able to develop their research abilities and master the most cutting-edge developments in mathematics.
PO-6	The students will be able to identify different sorts of research, its goals, write a thesis, and create a mathematical document in Latex.

10. Eligibility for admission

A candidate who has passed the undergraduate course like M.Sc., Mathematics / M.Sc., Mathematics (Computer Applications) degree of this University or any of the above degree of any other University shall be eligible for admission in Master of Philosophy (M.Phil.,) Degree in Mathematics of this University.

11. Medium of Instruction

English

12. Minimum Duration of programme

The programme is for a period of one year. Each year shall consist of two semesters viz. Odd and Even semesters. Odd semesters shall be from June / July to October / November and even semesters shall be from November / December to April / May. Each Semester there shall be 90 working days consisting of 6 teaching hours per working day (5 days/week).

- **Plan of work**

The candidate shall undergo Dissertation Work during the fourth semester. The candidate shouldprepare a scheme of work for the dissertation and should get approval from the guide. The candidate, after completing the dissertation work, shall be allowed to submit to the university at the end of the fourth semester. If the candidate is desirous of availing the facility from other universities/laboratory, they will be permitted only after getting approval from the guide. In such case, the candidate shall acknowledge the same in their dissertation.

Project/Dissertation

The candidate shall undergo Project/Dissertation Work during the final semester. The candidate should prepare a scheme of work for the dissertation/project and should get approval from the guide. The candidate, after completing the dissertation /project work, shall be allowed to submit it to the university departments at the end of the final semester. If the candidate is desirous of availing the facility from other departments/universities/laboratories/organizations they will be permitted only after getting approval from the guide and HOD. In such a case, the candidate shall acknowledge the same in their dissertation/project work.

- **Format to be followed for dissertation/project report**

The format /certificate for thesis to be followed by the student are given below

- Title page
- Certificate
- Acknowledgment
- Content as follows:

Chapter No	Title	Page number
1	Introduction	
2	Aim and objectives	
3	Review of literature	
4	Materials and methods	
5	Result	
6	Discussion	
7	Summary	
8	References	

- **Format of the title page**

Title of Dissertation/Project work

Dissertation/Project submitted in partial fulfilment of the requirement for the degree of Master of Science to the Alagappa University, Karaikudi -630003.

By

(Student Name)

(Register Number)

University Logo

Department of ----------

Alagappa University

(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the ThirdCycleand Graded as Category-I University by MHRD-UGC, 2019: QS ASIA Rank-216, QS BRICS Rank-104,QS India Rank-20)

Karaikudi - 630003

(Year)

- **Format of certificates**

Certificate -Guide

This is to certify that the **Dissertation/Project** entitled "----------" submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the degree of Master of Science in ---------- by Mr/Mrs ----------(Reg No----------) under my supervision. This is based on the results of studies carried out by him/her in the Department of----------, Alagappa University, Karaikudi-630 003. This Dissertation/Project or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar titles or record of any University or Institution.

Research Supervisor

Place:Karaikudi
Date:_____

Certificate - (HOD)

This is to certify that the thesis entitled **" ---------- "** submitted by Mr/Mis ----------(Reg No: ----------) to the Alagappa University, in partial fulfilment for the award of the degree of Master of --------in ---------- is a bonafide record of research work done under the supervision of Dr.----------, AssistantProfessor, Department of----------, Alagappa University. This is to further certify that the thesis or any part thereof has not formed the basis of the award to the student of any degree, diploma, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi

Head of the Department

Date:_____

Declaration (student)

I hereby declare that the dissertation entitled **" ----------"** submitted to the Alagappa University for the award of the degree of Master of -------- in ---------- has been carried out by me under the guidance of Dr. ----------, Assistant Professor, Department of ----------, Alagappa University, Karaikudi – 630 003. This is my original and independent work and has not previously formed the basis of the award of any degree, diploma, associateship, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi (_____)

Date:_____

Internship

The students shall undergo Internship / industrial training in the reputed organizations for minimum of two weeks to acquire industrial knowledge during the summer vacation of second semester. The students have to find industry related to their discipline (Public limited/Private Limited/owner/NGOs etc.,) in consultation with the faculty in charge/Mentor and get approval from the Head of the Department and Departmental Committee before going for an internship / industrial training.

- **Format to be followed for Internship report**

The format /certificate for internship report to be followed by the student are given below

- Title page -**Format of the title page**

Title of internship report

Internship report submitted in partial fulfilment of the requirement for the Master of degree in ---------- to the Alagappa University, Karaikudi -630003.

By

(Student Name)

(Register Number)

University Logo

Department of ----------

Alagappa University

(A State University Accredited with "A+" grade by NAAC (CGPA: 3.64) in the ThirdCycle and Graded as Category-I University by MHRD-UGC, 2019: QS ASIA Rank- 216, QS BRICS Rank-104,QS India Rank-20)

Karaikudi – 630003

(Year)

- Certificate-(**Format of certificate – faculty in-charge)**

This is to certify that the report entitled " ---------- " submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the Master of Science in ----------by Mr/Mis---------- (Reg No----------) under my supervision. This is based on the work carried out by him/her in the organization M/S----------. This Internship report or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar record of any University or Institution.

Place: Karaikudi Research Supervisor

Date:_____

Certificate (HOD)

This is to certify that the Internship report entitled **" ---------- "** submitted by Mr/Mis**.----------(Reg No ----------)** to the Alagappa University, in partial fulfilment for the award of the Master of Science in-------- is a bonafide record of Internship report done under the supervision of----------, Assistant Professor, Department of---------- , Alagappa University and the work carried out by him/her in the organization M/S ----------. This is to further certify that the thesis or any part thereof has not formed the basis of the award to the student of any degree, diploma, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi Head of the Department

Date:_____

- Certificate-(**Format of certificate – Company supervisor or Head of theOrganization)**

This is to certify that the Internship report entitled "----------" submitted to Alagappa University, Karaikudi-630 003 in partial fulfilment for the Master of Science in ----------by Mr/Mis---------- (Reg No ----------) under my supervision. This is based on the work carried out by him/her in our organization M/S --------- ---------- for the period of three months or ------. This Internship report or any part of this work has not been submitted elsewhere for any other degree, diploma, fellowship, or any other similar record of any University or Institution.

Place: Karaikudi Supervisor in charge

Date:_____

Declaration (student)

I hereby declare that the Internship Report entitled **" ---------- "** submitted to the Alagappa University for the award of the **Master of Science in** ----------has been carried out by me under the supervision of ---------- , Assistant Professor, Department of----------, Alagappa University, Karaikudi – 630 003. This is my original and independent work carried out by me in the organization M/S ---------- for the period of three months or------- and has not previously formed the basis of the award of any
degree, diploma, associateship, fellowship, or any other similar title of any University or Institution.

Place: Karaikudi (_____)

Date:_____

- Acknowledgment
- Content as follows:

Chapter No	Title	Page number
1	Introduction	
2	Aim and objectives	
3	Organisation profile /details	
4	Methods / Work	
5	Observation and knowledge gained	
6	Summary and outcome of the Internship study	
7	References	

- **No. of copies of the dissertation/internship report**

The candidate should prepare three copies of the dissertation report and submit the same for the evaluation of examiners. After evaluation, one copy will be retained in the department library, one copy will be retained by the guide and the student shall hold one copy. The candidate should prepare one copy of the field visit/internship report and submit the same for the evaluation of examiners.

13. Teaching methods

The method of teaching is by giving lectures, tutorials, seminars and supervised research projects. Moreover, extensive use is made of IT and a wide range of materials is available to enable students to study at their own place and in their own time to enhance and extend the material taught formally.

14. Attendance

Students must have earned 75% of attendance in each course for appearing for the examination. Students who have earned 74% to 70% of attendance need to apply for condonation in the prescribed form with the prescribed fee. Students who have earned 69% to 60% of attendance need to apply for condonation in the prescribed form with the prescribed fee along with the Medical Certificate. Students who have below 60% of attendance are not eligible to appear for the End Semester Examination (ESE). They shall re- do the semester(s) after completion of the programme.

15. Examination

The examinations shall be conducted separately for theory and practicals to assess (remembering, understanding, applying, analysing, evaluating, and creating) the knowledge required during the study. There shall be two systems of examinations viz., internal and external examinations. The internal examinations shall be conducted as Continuous Internal Assessment tests I and II (CIA Test I & II).

A. Internal Assessment

The internal assessment shall comprise a maximum of 25 marks for each subject. The following procedure shall be followed for awarding internal marks.

Theory -25 marks

Sr.No	Content	Marks
1	Average marks of two CIA test	15
2	Attendance	2
3	Seminar/group discussion/quiz	4
4	Assignment/field trip report/case study report	4
	Total	25

Practical -25 Marks

1	Major Experiment	10 marks
2	Minor Experiment	5 marks
3	Spotter (2x 5/ 4 x4) or any other mode	10 marks
	Total	25 Marks

Project/Dissertation/internship-50 Marks (assess by Guide/incharge/HOD/supervisor)

1	Two presentations (mid-term)	30 Marks
2	Progress report	20 Marks
	Total	50 Marks

B. External Examination

- There shall be examinations at the end of each semester, for odd semesters in the month of October / November; for even semesters in April / May.
- A candidate who does not pass the examination in any course(s) may be permitted to appear in such failed course(s) in the subsequent examinations to be held in October / November or April / May. However candidates who have arrears in Practical shall be permitted to take their arrear Practical examination only along with Regular Practical examination in the respective semester.
- A candidate should get registered for the first semester examination. If registration is not possible owing to shortage of attendance beyond condonation limit / regulation prescribed OR belated joining OR on medical grounds, the candidates are permitted to move to the next semester. Such candidates shall re-do the missed semester after completion of the programme.
- For the Project Report/ Dissertation Work / internship the maximum marks will be 150 marks for project report evaluation and for the Viva-Voce it is 50 marks (if in some programmes, if the project is equivalent to more than one course, the project marks would be in proportion to the number of equivalent courses).
- Viva-Voce: Each candidate shall be required to appear for Viva-Voce Examination (in defense of the Dissertation Work /Project/ internship).

C. Scheme of External Examination (Question Paper Pattern)

Theory - Maximum 75 Marks

Section A	10 questions. All questions carry equal marks. (Objective type questions)	10 x 1 = 10 Marks	10 questions – 2 each from every unit
Section B	5 questions Either / or type like 1.a (or) b. All questions carry equal marks and each answer should not exceed one page or 250 words.	5 x 5 = 25	5 questions – 1 each from every unit
Section C	Essay type questions 5 out of 8 questions. All questions carry equal marks and each answer should not exceed two page.	5 x 8 = 40	8 questions – Should cover all units

Practical –Maximum 75 Marks

Section A	Major experiment	15 Marks
Section B	Minor experiment	10 Marks
Section C	Experimental setup	5 Marks
Section D	Spotters (5 x 5 marks)	25 Marks
Section E	Record note	10 Marks
Section F	Viva voce	10 Marks

16. Dissertation /Project report/Internship report Scheme of evaluation

Dissertation /Project report/Internship report	150 Marks
Viva voce	50 Marks

17. Results

The results of all the examinations will be published through the Department where thestudent underwent the course as well as through University Website

18. Passing minimum

- A candidate shall be declared to have passed in each course if he/she secures not less than 40% marks in the End Semester Examinations and 40% marks in the Internal Assessment and not less than 50% in the aggregate, taking Continuous assessment and End Semester Examinations marks together.
- The candidates not obtained 50% in the Internal Assessment are permitted to improve their Internal Assessment marks in the subsequent semesters (2 chances will be given)by writing the CIA tests and by submitting assignments.
- Candidates, who have secured the pass marks in the End-Semester Examination andin the CIA but failed to secure the aggregate minimum pass mark (E.S.E + C I.A), are permitted to improve their Internal Assessment mark in the following semester and/or in University examinations.
- A candidate shall be declared to have passed in the Project / Dissertation / Internshipif he /she gets not less than 40% in each of the Project / Dissertation / Internship Report and Viva-Voce and not less than 50% in the aggregate of both the marks for Project Report and Viva-Voce.
- A candidate who gets less than 50% in the Project / Dissertation / Internship Report must resubmit the thesis. Such candidates need to take again the Viva-Voce on the resubmitted Project report.

20. Grading of the Courses

The following table gives the marks, Grade points, Letter Grades and classifications meant to indicate the overall academic performance of the candidate.

Conversion of Marks to Grade Points and Letter Grade (Performance in Paper / Course)

RANGE OF MARKS	GRADE POINTS	LETTER GRADE	DESCRIPTION
90 - 100	**9.0 – 10.0**	**O**	**Outstanding**
80 - 89	**8.0 – 8.9**	**D+**	**Excellent**
75 - 79	**7.5 – 7.9**	**D**	**Distinction**

70 - 74	**7.0 – 7.4**	**A+**	**Very Good**
60 - 69	**6.0 – 6.9**	**A**	**Good**
50 - 59	**5.0 – 5.9**	**B**	**Average**
00 - 49	**0.0**	**U**	**Re-appear**
ABSENT	**0.0**	**AAA**	**ABSENT**

a) Successful candidates passing the examinations and earning GPA between 9.0 and 10.0 and marks from 90 – 100 shall be declared to have Outstanding (O).

b) Successful candidates passing the examinations and earning GPA between 8.0 and 8.9 and marks from 80 - 89 shall be declared to have Excellent (D+).

c) Successful candidates passing the examinations and earning GPA between 7.5 – 7.9 and marks from 75 - 79 shall be declared to have Distinction (D).

d) Successful candidates passing the examinations and earning GPA between 7.0 – 7.4 and marks from 70 - 74 shall be declared to have Very Good (A+).

e) Successful candidates passing the examinations and earning GPA between 6.0 – 6.9 and marks from 60 - 69 shall be declared to have Good (A).

f) Successful candidates passing the examinations and earning GPA between 5.0 – 5.9 and marks from 50 - 59 shall be declared to have Average (B).

g) Candidates earning GPA between 0.0 and marks from 00 - 49 shall be declared to have Re-appear (U).

h) Absence from an examination shall not be taken as an attempt.

From the second semester onwards the total performance within a semester and continuous performance starting from the first semester are indicated respectively **by Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA).** These two are calculated by the following formulate

GRADE POINT AVERAGE (GPA) = $\Sigma_i C_i \; G_i / \Sigma_i C_i$

$$\text{GPA} = \frac{\text{Sum of the multiplication of Grade Points by the credits of the courses}}{\text{Sum of the credits of the courses in a Semester}}$$

21. Classification of the final result

CGPA	Grade	Classification of Final Result
9.5 – 10.0 9.0 and above but below 9.5	**O+** **O**	First Class – Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	**D++** **D+** **D**	First Class with Distinction*

7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	**A++** **A+** **A**	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	**B+** **B**	Second Class
0.0 and above but below 5.0	**U**	Re-appear

The final result of the candidate shall be based only on the CGPA earned by the candidate.

a) Successful candidates passing the examinations and earning CGPA between 9.5 and 10.0 shall be given Letter Grade (O+), those who earned CGPA between 9.0 and 9.4 shall be given Letter Grade (O) and declared to have First Class –Exemplary*.

b) Successful candidates passing the examinations and earning CGPA between 7.5 and 7.9 shall be given Letter Grade (D), those who earned CGPA between 8.0 and 8.4 shall be given Letter Grade (D+), those who earned CGPA between 8.5 and 8.9 shall be given Letter Grade (D++) and declared to have First Class with Distinction*.

c) Successful candidates passing the examinations and earning CGPA between 6.0 and 6.4 shall be given Letter Grade (A), those who earned CGPA between 6.5 and 6.9 shall be given Letter Grade (A+), those who earned CGPA between 7.0 and 7.4 shall be given Letter Grade (A++) and declared to have First Class.

d) Successful candidates passing the examinations and earning CGPA between 5.0 and 5.4 shall be given Letter Grade (B), those who earned CGPA between 5.5 and 5.9 shall be given Letter Grade (B+) and declared to have passed in Second Class.

i) Candidates those who earned CGPA between 0.0 and 4.9 shall be given Letter Grade (U) and declared to have Re-appear.

e) Absence from an examination shall not be taken as an attempt.

CUMULATIVE GRADE POINT AVERAGE (CGPA) = $\Sigma_n \Sigma_i C_{ni} \; G_{ni} / \Sigma_n \Sigma_i C_{ni}$

$$\text{CGPA} = \frac{\text{Sum of the multiplication of Grade Points by the credits of the entire Programme}}{\text{Sum of the credits of the courses for the entire Programme}}$$

Where '**Ci**' is the Credit earned for Course i in any semester; '**Gi**' is the Grade Point obtained by the student for Course i and 'n' refers to the semester in which such courses were credited.
CGPA (Cumulative Grade Point Average) = Average Grade Point of all the Courses passed starting from the first semester to the current semester.

Note: * The candidates who have passed in the first appearance and within the prescribed Semesters of the PG Programme are alone eligible for this classification.

19. Maximum duration of the completion of the programme

The maximum period for completion of **M.Sc.,** in **Mathematics** shall not exceed eight semesters continuing from the first semester.

20. Conferment of the Master's Degree

A candidate shall be eligible for the conferment of the Degree only after he/ she has earned the minimum required credits for the Programme prescribed therefor (i.e. 90 credits). Programme).

21. Village Extension Programme

The Sivaganga and Ramnad districts are very backward districts where a majority of people Lives in poverty. The rural mass is economically and educationally backward. Thus the aimof the introduction of this Village Extension Programme is to extend out to reach environmental awareness, social activities, hygiene, and health to the rural people of this region. The students in their third semester have to visit any one of the adopted villages within the jurisdiction of Alagappa University and can arrange various programs to educate the rural mass in the following areas for three day based on the theme.1. Environmental awareness 2. Hygiene and Health. A minimum of two faculty members can accompany the students and guide them.

ALAGAPPA UNIVERSITY, KARAIKUDI

Choice-based Credit system (CBCS)

(For the candidates admitted from the academic year 2022-2023 onwards)

M. Phil. Mathematics

S.No.	Course Code	Name of the course	Credits	Marks		
				Int.	Ext.	Total
SEMESTER-I						
1.	571101	Research Methodology	4	25	75	100
2.	571102	Measure Theory	4	25	75	100
3.	571103	General Skills in Geometry	4	25	75	100
SEMESTER-II						
4.	--	Specialization Course-I	4	25	75	100
5.	571999	Dissertation &Viva voce	8	Viva voce (50) + Dissertation (150)		200
		Total Marks	**24**	**--**		**600**

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Core Course	
Course Code	**Course Name**
571101	Research Methodology
571102	Measure Theory
571103	General Skills inGeometry
Core Course - Specialization	
Course Code	**Course Name**
571201	Functional Analysis
571202	Commutative Algebra
571203	Domination in Graphs
571204	Fractional Differential Equations

Semester – I			
Course Code: 571101	**Research Methodology**	**Credits: 4**	**Hours: 5**

Objectives	➢ Understanding the problem to be studied and identifying the research methodology of the relevant research area. ➢ Studying and identifying the MATLAB Software. ➢ Acquire the knowledge of MATLAB to use in Mathematical Applications. ➢ Acquire the knowledge of Environments and document layout in Latex. ➢ Develop their skills in Latex.
Unit - I	**Research Methodology-** An Introduction-Meaning of Research - Objectives of Research - Motivation in Research - Types of Research- Research Approaches - Significance of Research - Research Methods versus Methodology - Research and Scientific Method - Importance of Knowing How Research is Done- Research Process - Criteria of Good Research.
Unit - II	**Basics of MATLAB :** MATLAB windows- online help- Input output- File types-Platform defendence-General commands-Intractive Computation: Matrics and vectors-matric and array operation-character strings-Special note on array operation-Command line Functions-using built-in functions and online-help-plating simple graphs.
Unit - III	**Applications:** Linear Algebra- curve fitting and Interpolation-Data Analysis and Statistics-Numerical Integration-Ordinary differential equations.
Unit - IV	**Commands and Environments:** Command names and arguments,-Environments- Declaration-Lengths- Special characters-Spaces and carriage returns. Document Layout and Organization: Document class,-Page style-Parts of the document-Changing font,Centering and indenting- Lists-Theorem-like declarations.
Unit - V	**Mathematical Formulae:** Mathematical environments,-Main elements of math mode- Mathematical symbols-Additional elements, Fine-tuning mathematics-Horizontal spacing, selecting font size in formulas-processing parts of a document, In-text references, Bibliographies.

Textbook(s):

Kothari, C. R., (1990). *Research Methodology: Methods and Techniques* (2nd ed.). New Age International Publishers.
Rudra Pratap.(2015). *Getting Started with MATLAB.*Printed in India by Thomsan Press (India Ltd) Oxford University Press.
Kopka, H., Daly, P.W. (2004). *A Guide to LATEX* (4th ed.). London: Addision–Wesley.

References:

Gurumani, N. (2010). *Thesis Writing and Paper Presentation*. MJP Scientific Publishers.
Daniel T. Valentine and Brian D. Hahn(2022). *Essential MATLAB for Engineers and Scientists*, (8th ed.) Academic Press.
Kottwitz, S. (2011). *Latex Beginners Guide*. Packt publishing.

Outcomes	After the successful completion of this course, the student will be able to: ➢ Review literature to understand how others have approached or dealt with the problem. ➢ Learn all features of MATLAB as a programming tool. ➢ Work with a MATLAB programming. ➢ Write valid LaTeX documents that can be typeset on either a local LaTeX installation. ➢ Know various special formatting commands, including those for mathematics, text formatting and tables.

Course Designed by
Dr. N. Anbazhagan
Professor & Head
Department of Mathematics

Semester – I			
Course Code: 571102	**Measure Theory**	**Credits: 4**	**Hours: 5**

Objectives	➢ To introduce the concept of abstract measures and measure integration. ➢ To derive classical Lebesgue measure and Lebesgue integration as particular cases**.** ➢ introduce Lebesgue spaces L^p as normed spaces. ➢ provide representation theorems and duals of L^p spaces. ➢ derive Fubini's theorem and to introduce convolution.
Unit - I	**Abstract Integration:** Set - Theoretic notations and terminology - The concept of measurability - Simple functions - Elementary properties of measures - Arithmetic in [0, ∞] - Integration of positive measure - Integration of complex functions - The role played by sets of measurable functions.
Unit - II	**Positive Borel Measures**: Vector spaces - Topological preliminaries - The Riesz representation theorem - Regularity properties of Borel measures - Lebesgue measure - Continuity properties of measurable functions.
Unit - III	**L^P spaces:** Convex functions and inequalities - The L^p spaces-Approximation by continuous functions.
Unit - IV	**Complex Measure:** Total variations - Absolute continuity - Consequences of the Radon - Nikodym theorem - Bounded linear functionals on Lp - The Riesz representation theorem.
Unit - V	**Integration on Product Measures:** Measurability on cartesian products-Product measures - The Fubini theorem - Completion of product measure – Convolution - Distribution functions.

Textbook(s):

Rudin, W. (2006). *Real and Complex Analysis* (3rd ed.). Mc Graw Hill.

References:

Donald Coln, L. (2013). *Measure Theory* (2nd ed.). Birkhauser.

Carlos Kubrushy, S. (2007). *Measure Theory*. Springer.

Paul Halmos, R. (1914). *Measure Theory*. Springer.

Outcomes	After the successful completion of this course, the student will be able to: ➢ Unify classical Lebesgue integration and classical summation. ➢ Provide examples for Banach spaces and their duals through Lebesgue spaces. ➢ Applications of continuous functions were elaborately studied using L^p-spaces approximation. ➢ Understand the consequences of Radon-Nikodym theorem and The Riesz representation theorem. ➢ Get the knowledge of product measures, Convolution and Distribution functions.

Course Designed by
Dr. R. Jeyabalan
Assistant professor
Department of Mathematics

Semester - I			
Course Code: 571103	**General Skills in Geometry**	**Credits: 4**	**Hours: 5**

Objectives	➢ This course provides all classical geometry in plan analytical geometry. ➢ This course includes straight lines, circles, and pairs of straight lines, parabolas, ellipse, and hyperbolas. ➢ Introduce Cartesian coordinates and polar coordinates. ➢ To study the structure all second order general equations are completely analyzed and characterized.
Unit - I	Introduction -Algebraic Results-Coordinates-Lengths of Straight Lines and Areas of triangles -Locus-Equation to a locus-The Straight line-Rectangular Coordinates-The Straight Line -Polar Coordinates and Oblique Coordinates.
Unit - II	Equations Representing Two or More Straight lines -Transformation of Coordinates-The Circle.
Unit - III	Conic Sections: THE PARABOLA-Equation to a tangent-Some properties of the parabola-Pole and polar-Diameters-Equations in terms of one variable - Loci connected with the parabola- Three normals passing through a given point-Parabola referred to two tangents as axes.
Unit - IV	THE ELLIPSE-Auxiliary circle and eccentric angle-Equation to a tangent-Some properties of the ellipse- Pole and polar-Conjugate diameters-Four normals through any point-Examples of loci.
Unit - V	THE HYPERBOLA-Asymptotes-Equation referred to the asymptotes as axes-one variable-Examples.

Textbook(s):

S.L. Loney (1895). *The elements of Coordinate Geometry (First Edition).* Macmillan and co., New York

References:

Hari Kishan(2006) ,*Coordinate Geometry of Three Dimensions (First Edition),* Atlantic Publishers, New Delhi

Robert J. T. Bell(1923), *An elementary treatise on Coordinate geometry of Three Dimensions (Second edition),* Macmillan and co., London

H.B. Fine and H.D. Thompson(1911),*Coordinate Geometry*, The Macmillan company, New York

Outcomes	After the successful completion of this course, the student will be able to: ➢ Solve exercise problems in analytical geometry. ➢ Develop skills in solving problems is analytic geometry. ➢ Think logically and completely, because they are trained to think about the logical steps involved in solving problems. ➢ Understand the history of geometry and common procedures involved in the development of geometry.

Course Designed by
Dr. R. Jeyabalan
Assistant Professor
Department of Mathematics

Specialization Course			
Course Code: 571201	**Functional Analysis**	**Credits: 4**	**Hours: 5**

Objectives	➢ Focus on the definition of Topological vector spaces. ➢ Aim to introduce Banach - Steinhaus theorem. ➢ Introducing Milman's theorem and Hahn- Banach theorem. ➢ To investigate the compact operators. ➢ To illustrate how general methods of Haar measure on a compact space can be used.
Unit - I	**Topological Vector Spaces:** Normed spaces - vector Spaces – topological spaces - topological vector spaces -invariance - types of topological vector spaces - separation properties - Linear mappings - finite dimensional spaces- **Metrization** - boundedness and continuity –Seminorms and Local Converxity- quotient spaces- Examples.
Unit - II	**Completeness:** Baire category - Baire's theorem-the Banach – Steinhaus Theorem - the open mapping theorem - the closed graph theorem - bilinear mapping.
Unit - III	**Convexity:** The Hahn - Banach theorems - weak topologies - the weak topology of a topological vector space-the weak* topology of a dual space-compact convex sets- extreme points- the Krein Milman theorem- Milman's theorem.
Unit - IV	**Duality in Banach Spaces:** the normed dual of a normed space-duality- the second dual of a Banach space-Annihilators-Duals of subspaces and of quotient spaces-adjoints-compact operators.
Unit - V	**Some Applications**-A Continuity therorem-Closed Subspaces of L^p –Spaces-The range of a vector-valued measure-A generalized Stone-Weierstrass theorem- Two interpolation theorems-Kakutani's fixed point theorem-Haar measure on compact groups-Uncomplemented subspaces.

Textbook(s):

Rudin, W. (2017). *Functional Analysis*. (2nd ed.), Tata McGraw – Hill, 424 pp.

References:

Bachman, G., and L. Narici. (2000). *Functional Analysis*. Dover Publications, 1058pp.

Jean – pierre Aubin .(2000). *Applied Functional Analysis*. (2nd ed.), John Wiley & Sons, 510pp.

Kantororich, L.V., and G.P. Akilov. (2014). *Functional Analysis*. (2nd ed.), Pergemon Press**,** 604pp.

Outcomes	After the successful completion of this course, the student will be able to: ➢ Understand better the definition of Topological vector spaces ➢ Understand the proof of Hahn- Banach theorem ➢ Know the basic theory of Milman's theorem ➢ Consolidate earlier knowledge of compact operators through applications. ➢ Understand the applications of Haar measure on a compact space

Course Designed by
Dr. B. Sundaravadivoo
Assistant Professor
Department of Mathematics

Specialization Course			
Course Code: 571202	**Commutative Algebra**	**Credit: 4**	**Hours: 5**
Objectives	➢ Introduce the concept of free module, projective modules, tensor products and Flat modules as the generalization of a vector space. ➢ Study different types of ideals on local rings, localization and its algebraic applications. ➢ Establish Noetherian modules, Primary decomposition and Artinian modules ➢ Discuss about Integral extensions and its closed domains. ➢ Demonstrate the Discrete valuation rings and Dedekind domains.		
Unit - I	**Free modules** - Projective modules - Tensor products - Flat modules.		
Unit - II	**Ideals** - Local rings - Localization - Applications.		
Unit - III	**Noetherian modules** - Primary decomposition - Artinian modules - Length of modules.		
Unit – IV	**Integral elements** - Integral extensions - Integral closed domains -Finiteness of integral closure.		
Unit - V	**Valuation rings** - Discrete valuation rings - Dedekind domains.		
Textbook(s): Gopalakrishnan, N.S. (2015). *Commutative Algebra* (2nd ed.). University press. **References:** George Kemper. (2011). *A Course in commutative Algebra*. Springer. Sharp, R.Y. (2000). *Steps in Commutative Algebra* (2nd ed.). Cambridge Press. Ernst Kunz. (1985). *Introduction to Commutative Algebra and Algebraic Geometry*, Birkhauser.			
Outcomes	After the successful completion of this course, the student will be able to: ➢ Allocate features to free modules and demonstrate variety of examples. ➢ Access properties implied by different ideals on local rings and localizations. ➢ Analyze the Noetherian modules and Artinian modules by giving some illustration ➢ Determine integral elements, integral extensions and finiteness of integral closure. ➢ Understand the basic concepts of discrete valuation rings.		

Course Designed by
Dr. J. Vimala
Assistant Professor
Department of Mathematics

Specialization Course			
Course Code: 571203	**Domination In Graphs**	**Credits: 4**	**Hours: 5**

Objectives	➢ Calculate the bounds on the domination number in terms of order and size. ➢ Discuss the bounds of various domination parameters in terms of degree. ➢ Develop the interesting problem of domination numbers of planar graphs with small diameter. ➢ Discuss the concept independent sets and irredundant set of the graph. ➢ Discuss the domination number which does not increase when the graph is modified by removing a vertex or an edge.
Unit - I	**Dominating Queens** – Dominating Sets in Graphs – Sets of Representatives – School Bus routing – Bounds In Terms Of Order – Bounds in terms of order, Degree and Packing - Bounds in terms of order and size.
Unit - II	**Bounds in terms of Degree**, Diameter and Girth – Bounds in terms of independence and covering –
Unit - III	**Domination,Independence & Irredundance-**Hereditary and Super hereditary properties – Independent sets – Dominating sets – Irredundant sets.
Unit - IV	**The domination chain** – Extension using maximality and Minimality. (Non-trivial proofs of results without proofs are excluded from the syllabus)
Unit - V	**Changing and Unchanging Domination:** Changing:Vertex removal(CVR)-Edge removal(CER)-Edge addition(CEA)-Unchanging: Vertex removal(UVR)-Edge removal(UER)-Edge addition(UEA).

Textbook(s):

Teresa W.Haynes, Stephen T.Hedetniemi,Peter J.Slater. (1998). *Fundamentals of Domination in Graphs*. Marcel Dekker Inc..

References:

Bock Boon Lim. (2000). *On the Dominating Number of a graph*. National University of Singapore, 290pp.

Shaohwiwang. (2016). *On Topological indices and Dominating Numbers of graph*. University of Mississippi, 320pp.

Balakrishnan, R., and K. Ranganathan. (2012). *A text Book of Graph Theory*. (2nd ed.), 292pp

Outcomes	After the successful completion of this course, the student will be able to: ➢ Demonstrate a thorough knowledge of the NP completeness of the Domination problem, Identify the total domination number of graphs. ➢ Solve the irredundant number and matching number, Prove results for hereditary and super hereditary properties. ➢ Find the neighborhood knockout number and replacement. ➢ Understand a thorough knowledge of extension using maximality and minimality. ➢ Examine the effects on domination when the graphs is modified by deleting a vertex or deleting or adding an edge.

Course Designed by

Dr. S. Amutha

Assistant Professor

Ramanujan Centre for Higher Mathematics

Specialization Course			
Course Code: 571204	**Fractional Differential Equations**	**Credits: 4**	**Hours: 5**

Objectives	➢ Focus on the definition of Special Functions ➢ Aim to introduce Different types of fractional derivatives and integrals ➢ Introducing Existence and Uniqueness theorem ➢ To investigate the short-memory principle ➢ To illustrate how general methods of fractional controllers can be used
Unit - I	Special functions of the fractional calculus: Gamma function - Mittag-Leffler function-Generalized-Mittag-Leffler function-Functions of the Mittag-Leffler Type.
Unit - II	**Fractional derivatives and Integrals:** Grunwald - Letnikov Fractional Derivatives Riemann - Liouville fractional derivatives - Some Other approaches - Sequential fractional derivatives - Left and right fractional derivatives -Properties of fractional derivatives - Laplace Transforms of fractional derivatives.
Unit - III	**Existence and uniqueness theorems:** Linear fractional differential equations - Fractional differential equation of a general form - Existence and Uniqueness theorem as a method of solution - Dependence of a solution on initial conditions. The Laplace Transform method: Standard fractional differential equations-Ordinary linear fractional differential equations.
Unit - IV	**Numerical solution of fractional differential equations:** Initial Conditions- Numerical solution - Examples of numerical solutions - The "Short -Memory" Principle in Initial Value Problems for fractional differential equations.
Unit - V	**Fractional – order systems and controllers:** Fractional – order systems and fractional - Order Controllers-Example - On Fractional- order system Identification.

Textbook(s):

Podulbny, I. (1999). *Fractional Differential Equations*. Academic Press.

References:

Kilbas, A.A., Srivastava, H.M., Trujillo, J.J. (2006). *Theory and Applications of Fractional Differential Equations* (1st ed.). Elsevier Publication.

Constantin, M., Gheorghe, D., Jose Antonio, T.M. (2019). *Introduction to Fractional Differential Equations* (1st ed.). Springer International Publishing.

Zhou, Y. (2016). *Basic Theory of Fractional Differential Equation* (2nd ed.). World Scientific.

Outcomes	After the successful completion of this course, the student will be able to: ➢ Understand better the definition of Special Functions. ➢ Understand the proof of Different types of fractional derivatives and integrals. ➢ Know the basic theory of Existence and Uniqueness theorem. ➢ Consolidate earlier knowledge of short-memory principle through applications. ➢ Understand the applications of fractional controllers.

Course Designed by
Dr. B. Sundaravadivoo
Assistant Professor
Department of Mathematics

CURRICULUM VITAE

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Educational Qualification

Ph. D. (Mathematics) – Madurai Kamaraj University, Madurai, India. (2002).

M. Phil (Mathematics) - Madurai Kamaraj University, Madurai, India(1996).

M.Sc. (Mathematics) - Cardamom Planters Association College, Bodinayakanur.(1995).

B. Sc (Mathematics) - Cardamom Planters Association College, Bodinayakanur.(1993).

P.G.D.C.A – ICC & CE of Madurai Kamaraj University, Madurai, India. (1996).

Administrative Posts held during the Service:

Chairperson, School of Mathematics, Alagappa University, Karaikudi. (22.06.2015 to till date).

Head, Department of Mathematics, Alagappa University, Karaikudi. (05.06.2015 to till date).

Professional Experience

Professor of Mathematics, Alagappa University, Karaikudi. (09.03.2013 to till date).

Associate Professor of Mathematics, Alagappa University, Karaikudi. (09.03.2010 to 08.03.2013).

Reader in Mathematics, Alagappa University, Karaikudi. (09.03.2007 to 08.03.2010)

Lecturer in Mathematics, Thiagarajar College of Engineering, Madurai.(26.09.2001 to 08.03.2007)

Teaching Experience: 20 years

Research Experience: 20 Years

Area of Research

Stochastic Modelling, Graph theory and Datamining.

Honours and awards

Best Paper Presentation Award (2019) in the "International Conference on Mathematical Engineering Applications for Sustainable Development", held during August 16-18, 2019, University of Malaya, Meeting room 18-1, Wisma R&D, Kualalampur, Malaysia.

Excellence Award (2018) from Association of Inventory Academicians and Practitioners, New Delhi. (23.12.2018).

Tamilnadu Scientist Award (TANSA 2017) from Tamilnadu State Council for Science and Technology, Chennai, INDIA.(27.12.2019).

Outstanding Researcher Award (2017) from International Institute of Organized Research (I2OR), Chandigarh, India.

Research Award (2014-16) from University Grants Commission (UGC), New Delhi, India

Shri P. K. Das Memorial Best Faculty Award (2013) from Nehru Group of Institutions, Coimbatore, INDIA.

Career Award for Young Teachers (2005) from All India Council for Technical Education (AICTE), New Delhi, INDIA.

Young Scientist Fellowship (2005) from Tamilnadu State Council for Science and Technology, Chennai, India.

Young Scientist Award (2004) from Department of Science and Technology (DST), New Delhi, INDIA.

Recent Publications

1. JT. Harikrishnan, K. Jeganathan, S. Selvakumar, N. Anbazhagan, Woong Cho, Gyanendra Prasad Joshi and Kwang Chul Son, Analysis of Stochastic M/M/c/N Inventory System with Queue-Dependent Server Activation, Multi-Threshold Stages and Optional Retrial Facility, Mathematics, 10(15):2682, 37 pages, July 2022, https://doi.org/10.3390/math10152682.(ISSN: 2227-7390, SCIE, Scopus and Web of Science indexed, Imapct factor 2.592.)

2. V. Vinitha, N. Anbazhagan, S. Amutha, K. Jeganathan, Bhanu Shrestha, Hyoung-Kyu Song, Gyanendra Prasad Joshi and Hyeonjoon Moon, Analysis of a Stochastic Inventory Model on Random Environment with Two Classes of Suppliers and Impulse Customers, Mathematics, 10(13):2235, 226 pages, June 2022, https://doi.org/10.3390/math10132235.(ISSN: 2227-7390, SCIE, Scopus and Web of Science indexed, Imapct factor 2.258.)

3. K. Jeganathan, M. Abdul Reiyas, S. Selvakumar, N. Anbazhagan, S. Amutha, Gyanendra Prasad Joshi, Duckjoong Jeon and Changho Seo, Markovian Demands on Two Commodity Inventory System with Queue-Dependent Services and an Optional Retrial Facility, Mathematics, 10(12):2046, 22 pages, June 2022, https://doi.org/10.3390/math10122046.(ISSN: 2227-7390, SCIE, Scopus and Web of Science indexed, Imapct factor 2.258.)

4. N. Anbazhagan, Gyanendra Prasad Joshi, R. Suganya, S. Amutha, V. Vinitha, Bhanu Shrestha, Queueing-inventory System of Two-commodity with Optional Demands of Customers and MAP Arrivals, Mathematics, 10(11):1801, 12 pages, May 2022, https://doi.org/10.3390/math10111801.(ISSN: 2227-7390, SCIE, Scopus and Web of Science indexed, Imapct factor 2.258.)

5. K. Jeganathan, S. Vidhya, R. Hemavathy, N. Anbazhagan, Gyanendra Prasad Joshi, Chanku Kang and Changho Seo, Analysis of M/M/1/N Stochastic QueueingInventory System with Dis-cretionary Priority Service and Retrial Facility, Sustainability, 14(10):6370, 29 pages, May 2022,https://doi.org/10.3390/su14106370.(ISSN: 2071-1050, SCIE, Scopus and Web of Science indexed, Imapct factor 3.251.)

Total Publications: 101, Total Citation: 852, h- index: 17, i10- index: 27.

CURRICULUM VITAE

Name : **Dr. J. Vimala**

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Educational Qualification

Ph. D (Mathematics) - Alagappa University, Karaikudi, India. (2007)

M.Sc. (Mathematics) -Alagappa University, Karaikudi, India. (2002)

B. Sc (Mathematics) -Sri Meenakshi College for Women, Madurai, India. (2000)

B. Ed - Annamalai University, Chithambaram, India.

PGDCA - Alagappa University, Karaikudi, India. (2004)

Professional Experience

Assistant Professor, Department of Mathematics , Alagappa University, Grade -2.
(May 2016 to till date)
Assistant Professor, Department of Mathematics , Alagappa University, Grade -1.
(May 2012 -2016)
Teaching Experience: 20 years
Research Experience: 15 Years

Area of Research

Algebra –Lattice Theory
Fuzzy Algebra
Fuzzy Hyper structures
Decision Theory
Soft computing

Honours & Awards

Women Researcher Award-IOSRD 2018 , International Research Awards in Engineering Science & Technology, Chennai.
Distinguished Woman in Science (Mathematics) VIWA 2017, Venus International Foundation, Chennai.
Best Researcher Award (Mathematics) IMRF 2016, International Multidisciplinary Research Foundation, Goa.
Vallal Alagappar Research Recognition Award – 2020, Alagappa University.

Recent Publications

1. S. Sabeena Begam, J. Vimala*, "Compositions on lattice ordered multi-fuzzy soft matrix and its simulated application in medical diagnosis", **Mathematics and Computers in Simulation,** volume 201,Issue C, pp-555 - 563 2021, **(SCI) I.F: 3.601**

2. D. Preethi, J. Vimala*, S. Rajareega, "A systematic study in the applications of fuzzy hyperlattice", **AIMS Mathematics,** Vol. 6, No. 2, 1695-1705, 2021, **(SCIE) I.F :1.427.**

3. D. Preethi, J. Vimala*, S. Rajareega, Madeline Al-Tahan, "Fuzzy Hyperlattice Ordered δ - Group and Its Application on ABO Blood Group System", **Journal of Intelligent & Fuzzy Systems,** Vol. 41, No. 5, 2021, pp. 5309-5315 **(SCIE) I.F:*1.851*.**

Total Publications: 50 Cumulative Impact Factor: 33.294 Total citations: 298 h- index : 10

CURRICULUM VITAE

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Educational Qualification

PhD (Mathematics) - Periyar University, Salem, India. (2011)
M. Phil (Mathematics) - Periyar University, Salem, India. (2006)
M.Sc (Mathematics) - Periyar University, Salem, India. (2005)
B. Sc (Mathematics) - Govt. Arts College Salem, Tamilnadu, India. (2003)

Professional Experience

Assistant Professor, RCHM, Alagappa University, Grade -2. (May 2016 to till date)
Assistant Professor, RCHM, Alagappa University, Grade -1. (May 2012 -2016)
Teaching Experience: 10 years
Research Experience: 8 Years

Area of Research

Abstract & Fractional Differential Equations
Stability Analysis of Dynamical Systems
Neural Networks
Synchronization Theory
Mathematical Modeling and Population Systems
Genetic Regulatory Networks, Complex Dynamical Networks, Multi-Agent Systems

Honours & Awards

2020, 325 years of Fractional Calculus Award for the year 2020 in testimony of the High regard of the achievements in the Area of Fractional Calculus and its Applications by Biruni University, Istanbul, Turkey.

2020, Vallal Alagappar Research recognition Award for the year 2020 on the contribution towards the enhancement of Research Outcome of the University in the form of h-index by Alagappa University, Karaikudi.

2018, Outstanding Reviewer Award for the year 2018 in Mathematics and Computers in Simulation (Elsevier).

2017, Outstanding Reviewer Award for the year 2017 in Journal of the Franklin Institute (Elsevier).

2015, Awarded travel grant from NBHM for attending ICIAM in Beijing, China.

2014, Awarded travel grant from NBHM for attending ICM in Seoul, South Korea.

2010, Received Sir. C.V. Raman Budding Innovator Award from Periyar University, Salem.

2008, Awarded Senior Research Fellow under Rajiv Gandhi National Fellowship, New Delhi.

2005, Awarded Junior Research Fellow under Rajiv Gandhi National Fellowship, New Delhi.

Recent Publications

1. J Dianavinnarasi, R Raja, J Alzabut, J Cao, M Niezabitowski, O Bagdasar, Application of Caputo–Fabrizio operator to suppress the Aedes Aegypti mosquitoes via Wolbachia: An LMI approach Mathematics and Computers in Simulation 201 (2022) 462-485.

2. A Pratap, R Raja, RP Agarwal, J Alzabut, M Niezabitowski, E Hincal, Further results on asymptotic and finite-time stability analysis of fractional-order time-delayed genetic regulatory networks, Neurocomputing 475 (2022) 26-37.

3. A Stephen, R Raja, J Alzabut, Q Zhu, M Niezabitowski, O Bagdasar, Mixed time-delayed nonlinear multi-agent dynamic systems for asymptotic stability and non-fragile synchronization criteria, Neural Processing Letters 54 (1) (2022) 43-74

4. SA Jose, R Raja, Q Zhu, J Alzabut, M Niezabitowski, VE Balas, Impact of strong determination and awareness on substance addictions: A mathematical modeling approach, Mathematical Methods in the Applied Sciences 45 (8) (2022) 4140-4160.

5. S Aadhithiyan, R Raja, J Alzabut, Q Zhu, M Niezabitowski, Robust non-fragile Mittag-Leffler synchronization of fractional order non-linear complex dynamical networks with constant and infinite distributed delays, Mathematical Methods in the Applied Sciences 45 (4) (2022) 2166-2189.

Total Publications: 107 Cumulative Impact Factor: 316.001 Total Citations: 2042

h-index: 28 i-10 index: 62.

CURRICULUM VITAE

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Educational Qualification

Ph. D (Mathematics), Alagappa University, Karaikudi, India (2018)

M. Phil (Mathematics), Madurai Kamaraj University, Madurai, India. (2009)

M.Sc. (Mathematics), CPA College, Madurai Kamaraj University, Madurai, India (1998)

B. Sc (Mathematics), CPA College, Madurai Kamaraj University, Madurai, India (1996)

CSIR-NET (Mathematics), (2010)

PGDOR (Operations Research), Pondicherry University, Puducherry, India (2004)

Professional Experience

Assistant Professor, Department of Mathematics , Alagappa University, Grade -1. (May 2012 -2017)

Assistant Professor, Department of Mathematics , Alagappa University, Grade -2. (May 2017 to till date)

Teaching Experience: 19 years

Research Experience: 4 Year

Area of Research

Differential Equations, Fractional Differential equations , Control Theory, Stability Theory

Optimal Control, Mathematical Modelling, Perturbation Theory.

Honours and awards

University IV th Rank Holder in M. Sc Mathematics, Madurai Kamaraj University, Madurai, India.

Recent Publications

1. B. Sundara Vadivoo G. Jothilakshmi, , Y. Almalki, A. Debbouche, M.Lavanya (September 2022), Relative Controllability analysis of multiple fractional order differential equations with multiple time delays, Applied Mathematics and Computation, https://doi.org/10.1016/j.amc.2022.127192 , ISSN: 0096-3003, **[SCI, Elsevier Publication, IF: 4.397]**

2. G. Jothilakshmi, B. S. Vadivoo, Y. Almalki, A. Debbouche (August 2022), Controllability analysis of multiple fractional order integro-differential damping systems with impulsive interpretation, Journal of Computational and Applied Mathematics, https://doi.org/10.1016/j.cam.2022.114204 , ISSN: 0377-0427, **[SCI, Elsevier Publication, IF: 2.872]**

Total Publications: 15 Cumulative Impact factor: 20 Total citations: 55

H-index: 4 i-10 index: 2.

CURRICULUM VITAE

Name : **Dr. S. Amutha**

Designation : Asssistant Professor

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Educational Qualification:

Ph. D (Mathematics) - Alagappa University, Karaikudi, India.(2011)

M. Phil (Mathematics) - Manonmaniam Sundaranar University, Tirunelveli, India.(2001)

M.Sc. (Mathematics) - Manonmaniam Sundaranar University, Tirunelveli, India. (1999)

B. Sc (Mathematics) - Sri Sarada College for Women, Tirunelveli, India, (1997).

Post Graduate Diploma in Computer Applications (P.G.D.C.A) - Manonmaniam Sundaranar University, Tirunelveli, India.(2000)

Professional Experience:

Assistant Professor, RCHM, Grade – 2. (May 2016 to till date).

Assistant Professor, RCHM, Grade – 1. (May 2012 – 2016).

Teaching Experience: 14 years

Research Experience: 10 Years

Area of Research:

Graph Theory, Domination Theory, Algorithmic Graph theory, Discrete Mathematics,

Cryptography and Mathematical modeling.

Honors and awards:

Bright Educator Award from International Institute of Organized Research for the year 2017.

Eminent Mathematician Award from The International Multidisciplinary Research Foundation for the year 2019.

Recent Publications:

1. K. Jeganathan , M. Abdul Reiyas , S. Selvakumar , N. Anbazhagan , S. Amutha , Gyanendra Prasad Joshi , Duckjoong Jeon and Changho Seo , Markovian Demands on Two Commodity Inventory System with Queue-Dependent Services and an Optional Retrial Facility, Mathematics , 10, 2046, 22 pages, June 2022.

2. N. Anbazhagan, Gyanendra Prasad Joshi, R. Suganya, S. Amutha, V. Vinitha, Bhanu Shrestha, Queueing-inventory System of Two-commodity with Optional Demands of Customers and MAP Arrivals, Mathematics, 10(11):1801, 12 pages, May 2022,

3. N. Anbazhagan, Srijana Acharya, V. Vinitha, S. Amutha, K. Jeganathan, Changho Seo, and Hyun-Il Kim, The MAP/PH/N/1, Queueing-Inventory System With Demands From a Random Environment, IEEE Access, 10, pp:47371-47383, May 2022.

4. K. Jeganathan, S. Selvakumar, S. Saravanan, N. Anbazhagan, S. Amutha, Woong Cho, Gyanendra Prasad Joshi and Joohan Ryoo, Performance of Stochastic Inventory System with a Fresh Item, Returned Item, Refurbished Item, and Multi-Class Customers, Mathematics, 10(7):1137, 37 pages, April 2022.

5. V. Vinitha, N. Anbazhagan, S. Amutha, K. Jeganathan, Gyanendra Prasad Joshi, Woong Cho and Suseok Seo, Steady State Analysis of Impulse customers and Cancellation policy in Queueing-Inventory System, Processes, 9(12):2146, 16 pages, November 2021.

6. K. Jeganathan, S. Selvakumar, N. Anbazhagan, S. Amutha and Porpattama Hammachukiattikul, “Stochastic modelling on M/M/1/N Inventory system with queue-dependent service rate and retrial facility”, AMS Mathematics, Vol.6, No.7 , 2021, pp. 7386-7420.

7. G. Muhiuddin, N. Sridharan D. Al-Kadi, S.Amutha and M. E. Elnair, “Reinforcement Number of a Graph with respect to Half-Domination”, Journal of Mathematics, Vol.2021, Article ID 6689816, 7 pages.

8. K. Jeganathan, T. Harikrishnan, S. Selvakumar, N. Anbazhagan, S. Amutha , Srijana Acharya, Rajendra Dhakal and Gyanendra Prasad Joshi, “Analysis of Interconnected Arrivals on Queueing-Iventory System with Two Multi-Server Service Channels and One Retrial Facility”, Electronics 2021, 10,576.

9. K. Suriya Prabha, S. Amutha, N. Anbazhagan and Ismail Naci Cangul, “Bounds on Coindependent Liar’s Domination in Graphs”, Journal of Mathematics, Vol.2021, Article ID 5544559, 6 pages.

Total Publications: 26, h-index: 5, i10 – index: 2.

CURRICULUM VITAE

Name : **Dr. R. Jeyabalan**
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Educational Qualification

Ph. D. (Mathematics) – Bharathidasan University, Trichy, India. (2015)

M. Phil. (Mathematics) – Alagappa University, Karaikudi, India. (2010)

M. Sc. (Mathematics) –Alagappa University, India.(2007)

B. Sc. (Mathematics) – Madurai Kamaraj University, India. (2005)

Professional Experience

Assistant Professor, Department of Mathematics , Alagappa University. (2016 to Till date)

Teaching Experience: 7 years

Research Experience: 8 Years

Area of Research

Magic Labeling Graph Theory

Fuzzy Magic Labeling Graph Theory

Fuzzy Topology

Recent Publications

1. **Split Domination in Neutrosophic Graphs**", M. Mullai , S. Broumi, R. Jeyabalan, R. Meenakshi Neutrosophic Sets and Systems ,Volume No. 47, Year 2021, Pages 240-249.

2. "**Consecutive z-index vertex magic labeling graphs**" R. Nishanthini, R.Jeyabalan, S. Balasundar and G.Kumar Journal of Intelligent and Fuzzy Systems , Volume Vol. 41, no. 1, Year 2021, Pages 219-230.

3. "**Strongly Edge Multiplicative Graphs**" R.Jeyabalan and G.Kumar Advances in Mathematics: Scientific Journal, Volume 1, Year 2020, Pages 173-185.

4. "**Strongly Vertex Multiplicative Graphs**" R.Jeyabalan and G.Kumar Advances in Mathematics: Scientific Journal, Volume 4, Year 2020, Pages 2177–2183.

5. "**A Single Valued Neutrosophic Inventory Model with Neutrosophic Random Variable**" M.Mullai and R.Jeyabalan and S. Broumi International Journal of Neutrosophic, Volume 1, Year 2020, Pages 47-57.

Total Publications**:** 12

CURRICULUM VITAE

Name : **Dr. M. Mullai**

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Alagappa University, Karaikudi

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Educational Qualification

Ph. D (Mathematics) – Alagappa University, Karaikudi, India. (2012)

M. Phil (Mathematics) – Alagappa University, Karaikudi, India.(2002)

M. Sc (Mathematics) – Alagappa University, Karaikudi, India.(2001).

B. Sc (Mathematics) – Alagappa Govt. Arts College, Karaikudi, (Madurai Kamaraj University), India. (1999).

P.G.D.C.A (Computer Applications) - Alagappa University, Karaikudi, India. (2001)

Professional Experience

Assistant Professor in Mathematics, Directorate of Distance Education, Alagappa University. (2013 – till date)

Associate Professor and Head, Department of Science and Humanities, Sri Raaja Raajan College of Engineering and Technology, Amaravathipudur. (2011 to 2013).

Assistant Professor, Department of Mathematics, St. Michael College of Engineering and Technology, Kalaiyarkoil. (2010 to 2011)

Lecturer, Department of Mathematics, Veltech, Avadi, Chennai. (2005 to 2006).

Guest Lecturer, Department of Mathematics, Alagappa Govt. Arts College, Karaikudi

(2004 to 2005 and from 2006 to 2010).

Teaching Experience: 18 years

Research Experience: 15 Years

Area of Research

Algebra & Fuzzy Algebra , Operations Research, Mathematical Modelling, Neutrosophic sets (Neutrosophic Inventory, Neutrosophic Graph theory, Neutrosophic Optimization, Neutrosophic Adhoc networks).

Honours and Awards

Best Article Award 2018 for the paper entitled Neutrosophic EOQ Model with price break by Neutrosophic Science International Association (NSIA), University of New Mexico, Maths and Science Department, 705, Gurley Ave, Gallup NM 87301, USA, 7 February 2019.

Recent Publications

1. M.Mullai., G.Madhan Kumar., Modeling social media addiction with case detection and treatment, Stochastic analysis and applications, 2022, 1-32, https://doi.org/10.1080/07362994.2022.2087677.

2. M. Mullai , S. Broumi., et.al., Inverse Dominating Set in Neutrosophic Graphs, International Journal of Neutrosophic Science (IJNS) Vol. 18, No. 3, PP. 104-110, 2022.

3. M. Mullai , S. Broumi., et.al., Equitable Domination in Neutrosophic Graphs, Journal of Neutrosophic and Fuzzy Systems (JNFS) Vol. 2, No. 2, PP. 57-61, 2022.

4. M. Mullai , S. Broumi., et.al., Split Domination in Neutrosophic Graphs, Neutrosophic Sets and Systems, Vol. 47, 2021.

5. M. Mullai and R. Surya., Neutrosophic Inventory Backorder Problem Using Triangular Neutrosophic Numbers, Neutrosophic Sets and Systems, Vol. 31, 2020.

Total Publications: 43, Total Citations: 172, h-index: 08, i10-index: 07